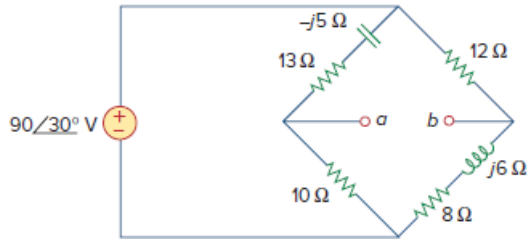


Problem

Find the Thevenin and Norton equivalent circuits at terminals a - b in the circuit of Fig. 10.110.

Figure 10.110:



Step-by-step solution

Step 1 of 11

Refer to the Figure 10.110 in the textbook.

Draw the circuit to calculate the Thevenin resistance.

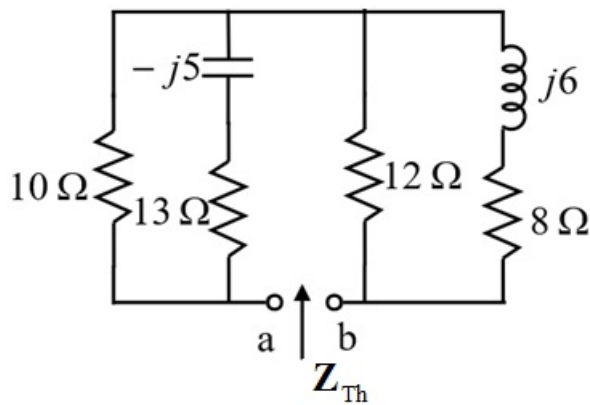


Figure 1

Step 2 of 11

The impedances $13 - j5 \Omega$ and 10Ω are connected in parallel.

Calculate the equivalent resistance between $13 - j5 \Omega$ and 10Ω .

$$\begin{aligned} Z_1 &= \frac{(13 - j5)(10)}{13 - j5 + 10} \\ &= \frac{130 - j50}{23 - j5} \\ &= 5.84 - j0.9 \Omega \end{aligned}$$

Step 3 of 11

The impedances $8 + j6 \Omega$ and 12Ω are connected in parallel.

Calculate the equivalent resistance between $8 + j6 \Omega$ and 12Ω .

$$\begin{aligned} Z_2 &= \frac{(8 + j6)(12)}{8 + j6 + 12} \\ &= \frac{96 + j72}{20 + j6} \\ &= 5.39 + j1.98 \Omega \end{aligned}$$

Step 4 of 11

Calculate the value of the Thevenin's impedance.

$$\begin{aligned} Z_{Th} &= Z_1 + Z_2 \\ &= 5.84 - j0.90 + 5.39 + j1.98 \\ &= 11.23 + j1.08 \Omega \end{aligned}$$

Thus, the Thevenin's impedance is $11.23 + j1.08 \Omega$.

Step 5 of 11

Draw the circuit to calculate the Thevenin's voltage.

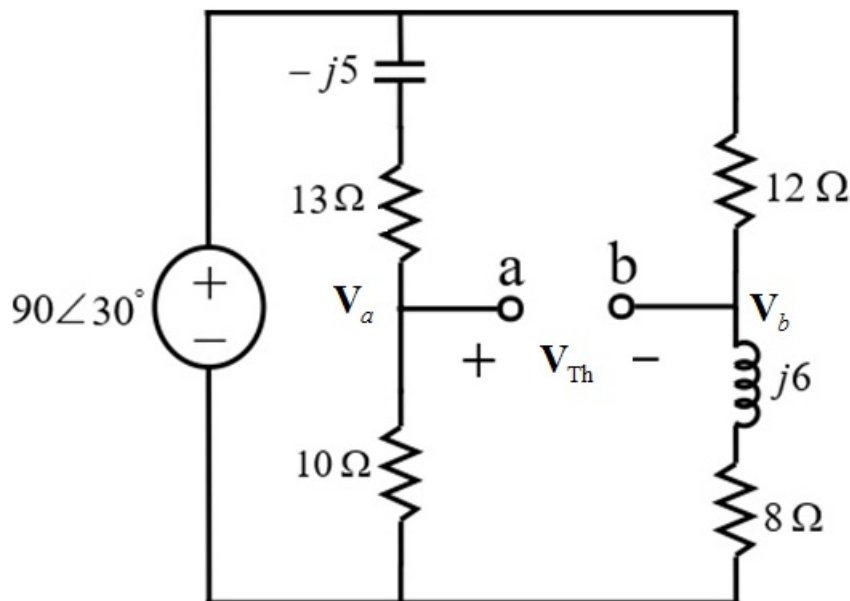


Figure 2

Step 6 of 11

Calculate the voltage, V_a .

$$\begin{aligned} V_a &= \left(\frac{10}{10 - j5 + 13} \right) (90\angle 30^\circ) \\ &= \frac{900\angle 30^\circ}{23 - j5} \\ &= 28.29 + j25.716 \text{ V} \end{aligned}$$

Step 7 of 11

Calculate the voltage, V_b .

$$\begin{aligned} V_b &= \left(\frac{8 + j6}{12 + j6 + 8} \right) (90 \angle 30^\circ) \\ &= \frac{(90 \angle 30^\circ)(8 + j6)}{20 + j6} \\ &= 27.607 + j33.1 \text{ V} \end{aligned}$$

Step 8 of 11

Calculate the value of Thevenin's voltage, V_{th} .

$$\begin{aligned} V_{th} &= V_a - V_b \\ &= (28.29 + j25.716) - (27.607 + j33.1) \\ &= 0.683 - j7.384 \\ &= 7.415 \angle -84.7^\circ \text{ V} \end{aligned}$$

Thus, the value of Thevenin's voltage is $7.415 \angle -84.7^\circ \text{ V}$.

Step 9 of 11

Draw the Thevenin's equivalent circuit.

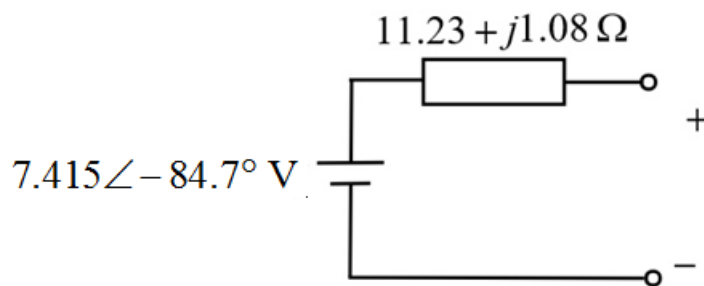


Figure 3

Thus, the Thevenin's equivalent circuit is shown in Figure 3.

Step 10 of 11

Calculate the value of Norton's current.

$$\begin{aligned} I_N &= \frac{V_{th}}{Z_{th}} \\ &= \frac{7.415 \angle -84.7^\circ}{11.23 + j1.08} \\ &= 0.6572 \angle -90.19^\circ \text{ A} \\ &= 657.2 \angle -90.19^\circ \text{ mA} \end{aligned}$$

Thus, the value of Norton's current is $657.2 \angle -90.19^\circ \text{ mA}$.

Step 11 of 11

Draw the Norton's equivalent circuit.

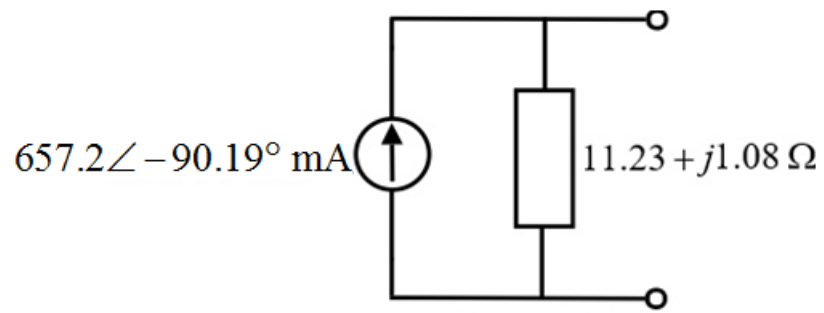


Figure 4

Thus, the Norton's equivalent circuit is shown in Figure 4.